

Serie 8

DIMENSION AND BASIS

1. Dimension. Compute the dimension and find a basis of the following spaces

- (a) The space of upper triangular matrices in $M_{n \times n}(\mathbb{R})$ over $\mathbb{R}$ (for a definition, see Serie 7, exercise 2.b));
- (b) The space of diagonal matrices in $M_{n \times n}(\mathbb{R})$ over $\mathbb{R}$, where diagonal matrices are all matrices $A = (a_{ij})$ such that $a_{ij} = 0$ whenever $i \neq j$;
- (c) The space of symmetric matrices

$$W = \{A \in M_{n \times n}(\mathbb{R}) \mid A^T = A\},$$

where $\cdot^T$ denotes the operation $A = (a_{ij})_{1 \leq i, j \leq n} \mapsto (a_{ji})_{1 \leq i, j \leq n}$;

- (d) The space of matrices $A \in M_{n \times n}(\mathbb{F}_2)$ such that the sum of the columns of A is the null vector.

2. Basis of a linear subspace. Let W be the linear subspace generated by the vectors

$$v_1 = \begin{pmatrix} 1 \\ 2 \\ 1 \\ 2 \end{pmatrix}, v_2 = \begin{pmatrix} 0 \\ 0 \\ -1 \\ 1 \end{pmatrix}, v_3 = \begin{pmatrix} 1 \\ 4 \\ 2 \\ 3 \end{pmatrix}, v_4 = \begin{pmatrix} 0 \\ 2 \\ 1 \\ 1 \end{pmatrix}.$$

- (a) Compute the dimension of W .
- (b) Determine a system of equations whose solution is W .

Hint. If the dimension of W is 3 and you find a vector n that is orthogonal to all $w \in W$, i.e. such that

$$0 = \left\langle \begin{pmatrix} n_1 \\ n_2 \\ n_3 \\ n_4 \end{pmatrix}, \begin{pmatrix} w_1 \\ w_2 \\ w_3 \\ w_4 \end{pmatrix} \right\rangle := \sum_{i=1}^4 n_i w_i$$

for all $w \in W$, then W is the solution to the system

$$n_1 x_1 + n_2 x_2 + n_3 x_3 + n_4 x_4 = 0.$$

- (c) Find all the possible bases of W that can be built using v_1, v_2, v_3, v_4 . How many are there?

3. Dimension. Determine a basis and the dimension of the following spaces:

(a) The set of solutions $S \subseteq \mathbb{R}^3$ of

$$\begin{aligned}x + y - z &= 0 \\3x + y + 2z &= 0 \\2x + 3z &= 0\end{aligned}$$

(b) $\{0\}$;

(c) $\{(z, w) \in \mathbb{C}^2 \mid z + iw = 0\}$ as a vector space over $\mathbb{C}$;

(d) $\{(z, w) \in \mathbb{C}^2 \mid z + iw = 0\}$ as a vector space over $\mathbb{R}$.

4. Dimension. Let K be a field, fix $g(X) := X + 5 \in K[X]$, and let $d \geq 1$. Compute the dimension of the space

$$W = \{h \in K[X] \mid \deg(h) \leq d \wedge \exists f \in K[X] : h = gf\}.$$

5. Dimension. Determine a basis and compute the dimension of the space

$$\{f : \mathbb{R} \rightarrow \mathbb{R} \mid f \text{ is twice differentiable} \wedge (f + f'' = 0)\}.$$

Hint: Note that f, f' , and f'' exist and are continuous. Moreover, you can use the following result from Analysis: the function $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $\forall x \in \mathbb{R} : f(x) = 0$ is the unique continuous solution to

$$\begin{cases} f + f'' &= 0 \\ f(0) &= f'(0) = 0 \end{cases}$$